Final

Quadrature Amplitude Modulation

Amplitude Shift keying (ASK)-

-> Advantages-

1. Very simple implementation.
2. Low cost and low power consumption.
3. Band efficient.

-> Disadvantages-

1. Highly susceptible to noise.
2. Not suitable for long distance or wireless communication.

-> Applications-

1. Optical fiber communication.
2. Short range data communication like sensor data.
3. IR remote control.

Phase shift keying (PSK)-

-> Advantages-

1. Better power efficiency than ASK & FSK.
2. Excellent noise immunity.
3. More bandwidth efficient.

-> Disadvantages-

1. Requires phase synchronization.
2. Sensitive to phase errors.
3. More complex transmitter and receiver.

—> Applications -

1. Wireless communication such as wifi, cellular system.
2. Satellite communication.
3. Digital broadcasting.

Frequency Shift Keying (FSK) -

—> Advantages -

1. Constant amplitude hence more efficient than ASK.
2. Less error prone.
3. Doesn't require phase synchronization.

—> Disadvantages -

1. Requires larger bandwidth.
2. Low data rate.
3. More complex than ASK.

—> Applications -

1. Telephone lines.
2. Low frequency systems.
3. Bluetooth.

* Best Flow :-

ASK < FSK < PSK

Feature

Feature	ASK	FSK	PSK
Basic Principle	Amplitude Carrier changes	Frequency Carrier Changes	Phase Carrier Changes
Noise immunity	Poor	Better than ASK	Best
Bandwidth	Low	High	Moderate
Power	Low	moderate	High
Complexity	Simple	Moderate	Complex

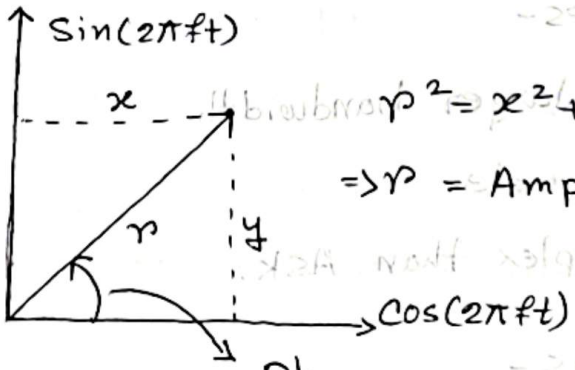
Quadrature amplitude modulation –

QAM is a modulation technique that combines ASK and PSK where both phase and amplitude of carrier changes while conveying the data signal.

A data point can be represented on constellation diagram.

Amplitude = Distance from origin.

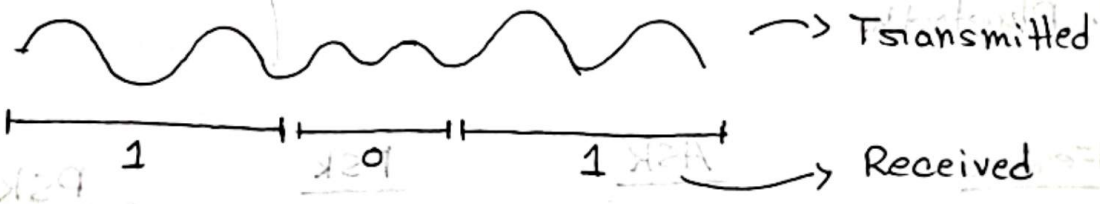
Phase = Angle with x-axis.

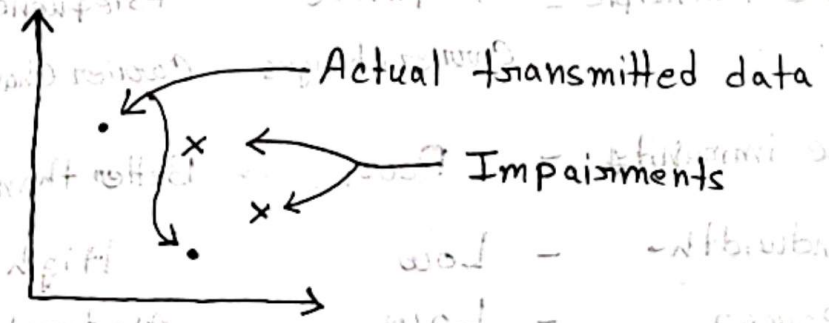


$r^2 = x^2 + y^2$

$\Rightarrow r$ = Amplitude

Phase = ϕ = A sin ϕ





-> Advantages -

1. Flexible and scalable.
2. Efficient use of bandwidth.
3. High efficiency and data rate.

-> Disadvantages -

1. Sensitive to noise.
2. Complex transmitter and receiver.
3. Requires power amplifiers.

-> Applications -

1. Wifi
2. 4G, 5G
3. Digital TV broadcasting.

Multiplexing

Multiplexing is a data communication method that combines multiple signals or data streams into one signal over a shared medium.

—> Enables efficient use of bandwidth.

—> Allows simultaneous transmission of multiple signals across a single data link.

3 considering factor -

1. Bandwidth:

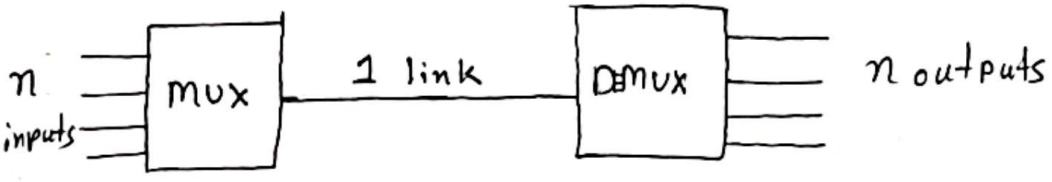
It is the range of frequencies a communication channel or signal occupies. It indicates the frequency space available to carry information.

2. Bitrate:

It signifies the number of bits transmitted per second.

3. Baudrate:

It is the number of signal changes or symbols sent per second over a communication channel.



Demultiplexing -

It is the process of separating combined signals or data streams back into their original individual channels at the receiver.

—> Bandpass filter - Separates different frequency signals.

—> Demodulator - Recovers original message.

Types of Multiplexing -

- —> Frequency Division Multiplexing (FDM) } Analog Multiplexing
- —> Wavelength Division Multiplexing (WDM) }
- —> Time Division Multiplexing (TDM) —> Digital Multiplexing
  - —> Synchronous TDM
  - —> Statistical TDM

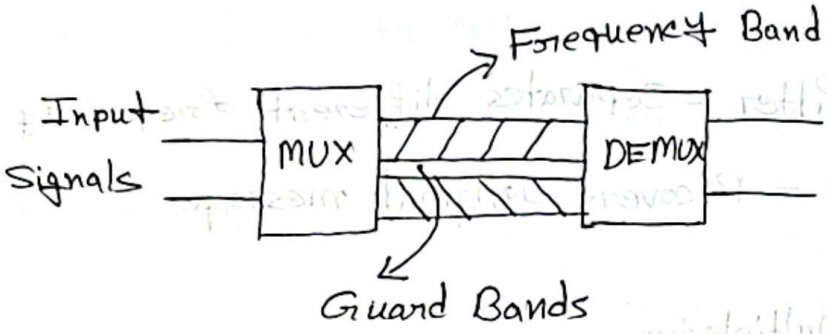
Frequency Division Multiplexing (FDM) -

A type of multiplexing method where the bandwidth of a single physical medium is divided into a number of smaller, independent frequency channels.

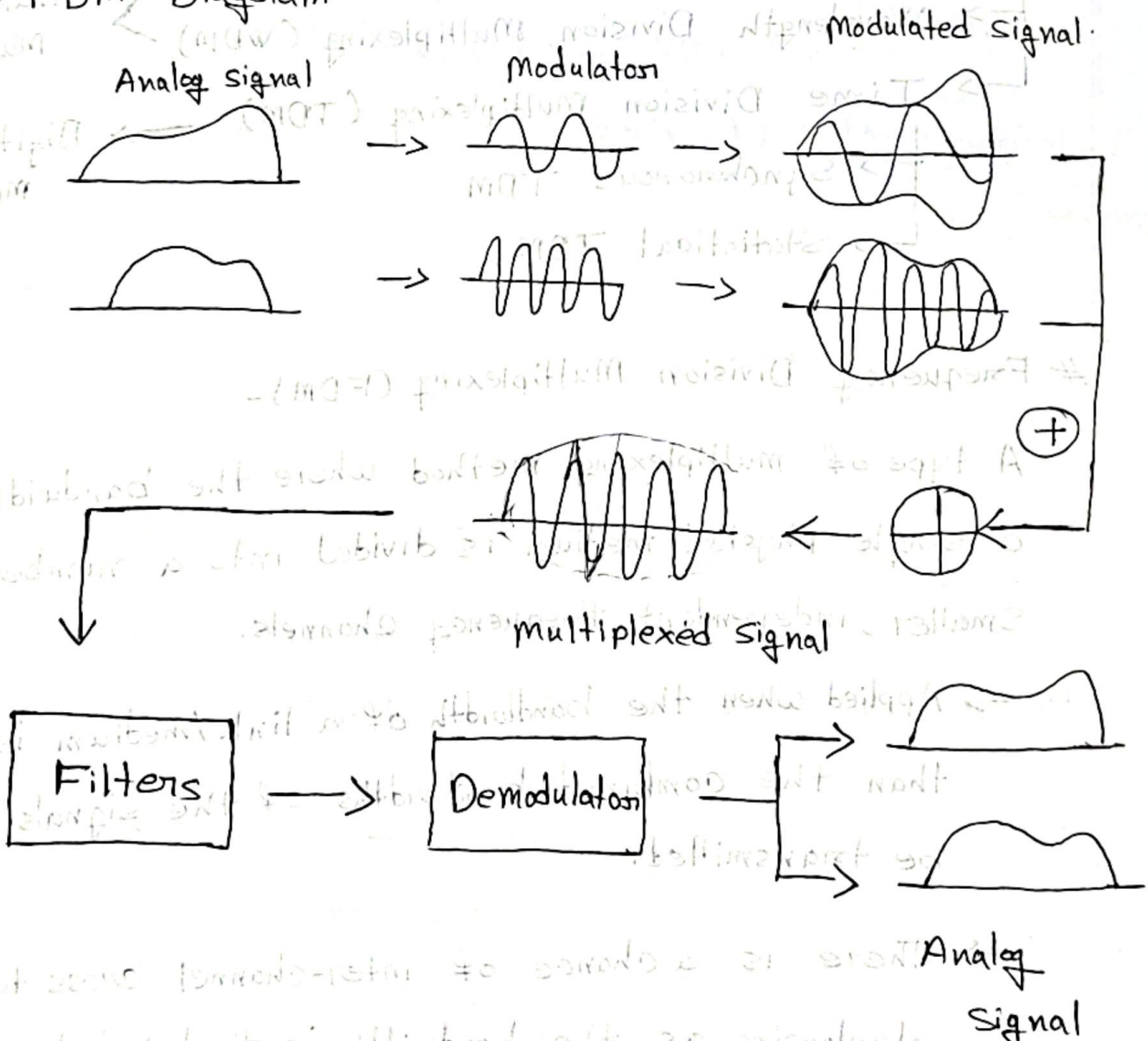
—> Applied when the bandwidth of a link/medium is greater than the combined bandwidths of the signals to be transmitted.

—> There is a chance of inter-channel cross talk ~~and overlapping~~ as the bandwidth is divided into multiple frequency channels.

-> To prevent the inter-channel cross talk and overlapping Unused strips of bandwidth are placed between each channel, these are called Guard bands.



FDM Diagram -



Prob 1

Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz. Let the channel bandwidth be from 20 to 32 kHz. Assuming these are no guard bands, show and explain the configuration.

=>

Given,

Each channel bandwidth = 4 kHz

Total channel = 3

Total available bandwidth = 12 kHz

Available frequence range = 20 kHz to 32 kHz

∴ Total bandwidth needed = 3 × 4 = 12 kHz

∵ Total bandwidth needed = ≤ Total available bandwidth

So, band need for transmission,

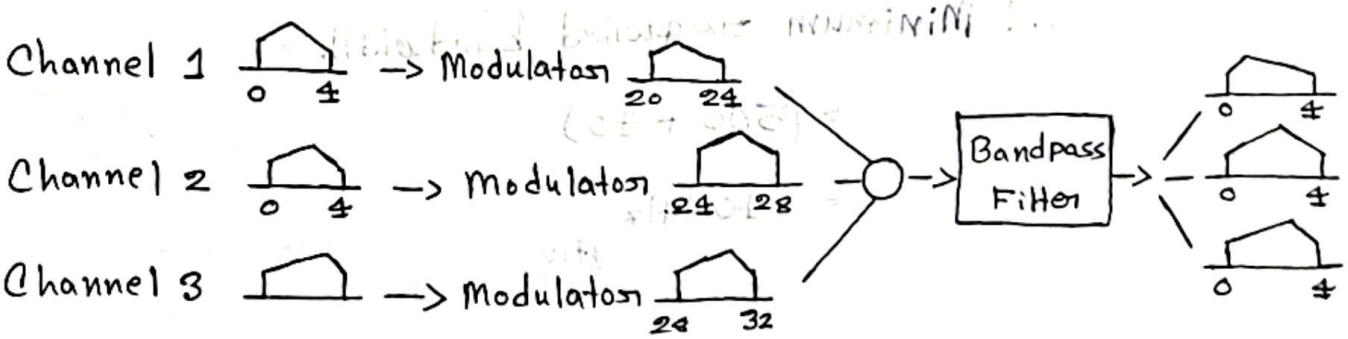
$$= \frac{32-20}{3} = \frac{12}{3} = 4 \text{ kHz each.}$$

So,

Channel 1 -> 20-24 kHz

Channel 2 -> 24-28 kHz

Channel 3 -> 28-32 kHz



Visualization

Prob 2

Assume 5 channels, each with a 100-kHz bandwidth, are to be multiplexed together. There is also need for guard band of 10 kHz between the channels to prevent overlapping. What is the minimum bandwidth of the link/medium ?

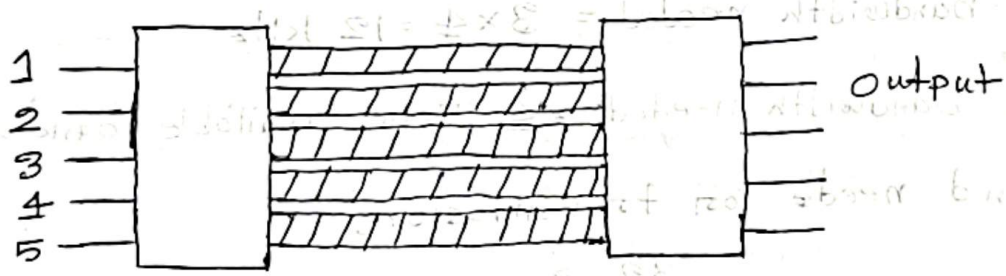
=>

Given,

Number of channels = 5

Bandwidth per channel = 100 kHz

Guard band between channels = 10 kHz.



∴ Total guard band = 5 - 1 = 4 guard bands

∴ Total Channel bandwidth = 5 x 100 = 500 kHz

∴ Total guard band bandwidth = 4 x 10 = 40 kHz

∴ Minimum required bandwidth

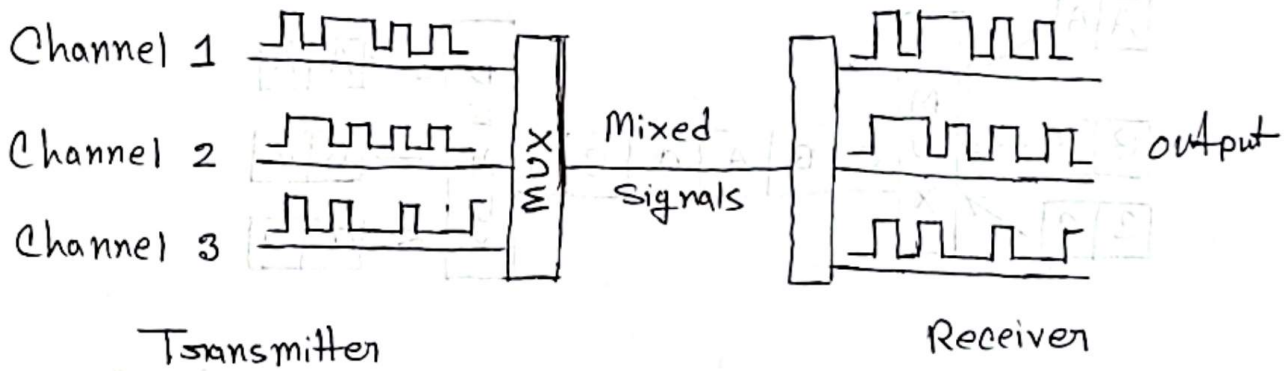
= (500 + 40)

= 540 kHz

Ans

Wavelength Division Multiplexing (WDM) -

A type of multiplexing method used to increase the capacity of optical fiber by transmitting multiple optical signals of different wavelengths simultaneously over a ~~signal~~ single optical fiber.



Types -

1. Dense WDM -

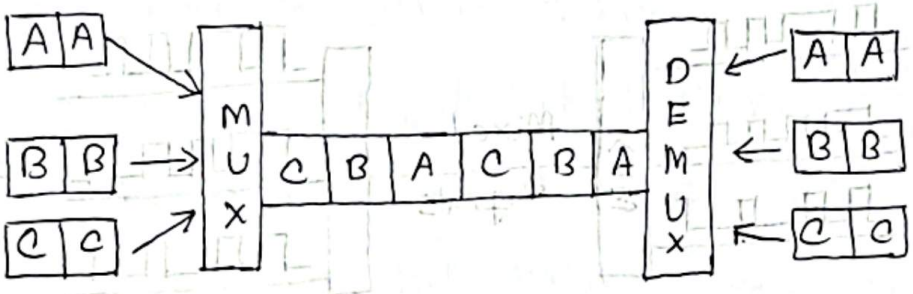
It is used to multiplex a large number of optical signals onto a single optical fiber. DWDM is used to carry different wavelengths that are very close to each other. It overcomes the speed and distance limitations of optical fiber and prevent attenuation as well.

2. Coarse WDM -

It is for lower-capacity applications compared to DWDM. The working principle is similar as it also combines different wavelengths to transmit over optical fiber, but can't send large amount of information

Time Division Multiplexing –

A type of multiplexing method of transmitting multiple signals over a single channel by dividing time into small intervals and allocating each signal a specific time slot in a repeating sequence.

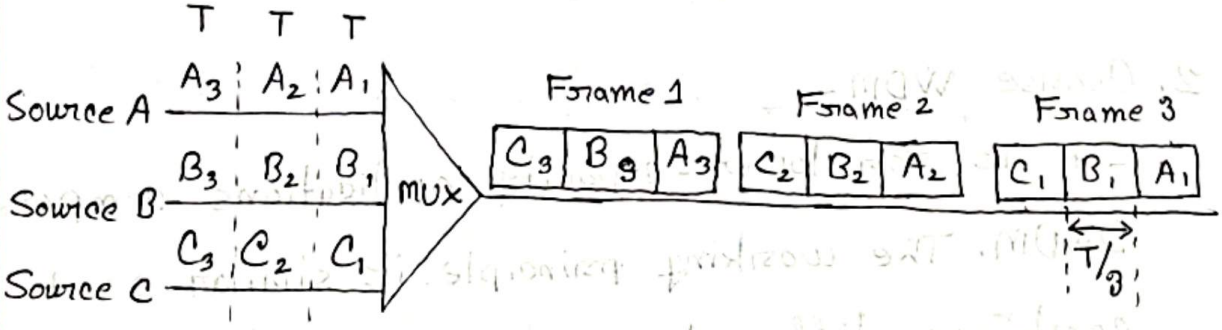


* Synchronous TDM –

-> Data divided into units called frame and each input have time slot to transmit data.

-> No. of source / channel = # No of slot in a frame.

-> Data taken at T interval, then the slot duration will be T/n.



(Inefficient)

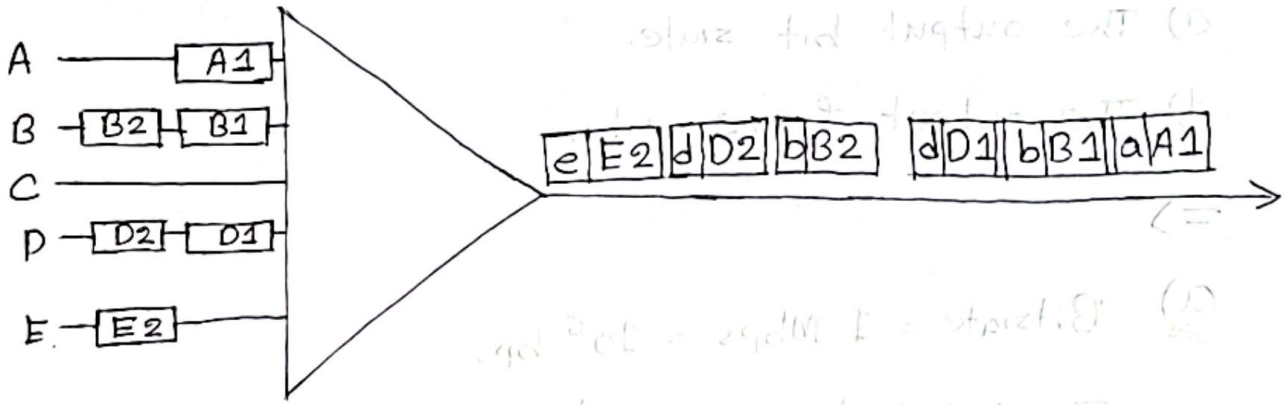
*Statistical TDM -

—> Slots are dynamically allocated to improve the bandwidth efficiency.

—> When source is ready to transmit data, it is given a slot in carrying frame.

—> A header or a destination address infront of the data.

—> Efficient as resources are not wasted.



*Orthogonal Frequency Division Multiplexing (OFDM) -

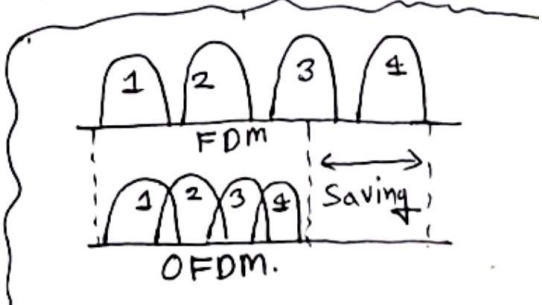
—> Here data is transmitted simultaneously over many closely spaced orthogonal subcarriers.

—> Used in digital system (Advance FDM).

—> Signals overlap each other unlike FDM, where band guard is necessary. to avoid cross talk.

—> Eliminates inter-channel cross talk.

—> Uses bandwidth more efficiently and provides high data right.



Ex - 6.6

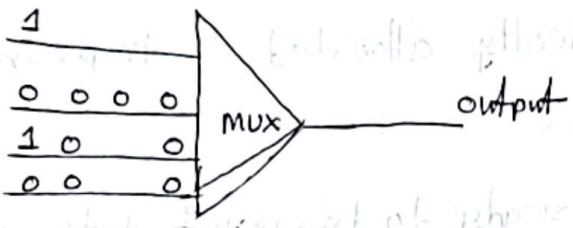


Fig shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit

a) Find the input bit duration.

b) The output bit duration.

c) The output bit rate.

d) The output frame rate.

=>

a) Bitrate = 1 Mbps = 10^6 bps

Input bit duration = $\frac{1}{10^6}$ = 1 μs

b) Output rate = 4 x 1 Mbps = 4 Mbps

Output bit duration = $\frac{1}{4 \times 10^6}$ = 0.25 μs

c) Output bit rate = 4 Mbps

d) Frame rate = 1×10^6 frames/sec

Ex 6.7 –

Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find–

a) The duration of 1 bit before multiplexing.

b) The transmission rate of the link.

c) The duration of a time slot.

d) The duration of a frame.

=>

a) Bit duration = $\frac{1}{1000}$ = 1 ms

Connection = 4
1 kbps = 1000 bps
Unit = 1 bit

b) Total rate = 4 x 1000 = 4000 bps = 4 kbps

c) Time slot duration = $\frac{1}{4000}$ = 0.25 ms
(1 bit at output rate)

d) Frame duration = 4 x 0.25 = 1 ms
(4 slot)

Ex 6.8

Four channel are multiplexed using TDM. If each channel sends 100 bytesis and we multiplex 1 byte per channel, show the frame travelling on the link, the size of the frame, the duration of a frame, the frame rate and the bit rate for the link.

=>

Frame stucture -

[Ch1: 1 byte][Ch2: 1 byte][Ch3: 1 byte][Ch4: 1 byte]

Frame size = 4 bytes = 32 bits

Frame rate = 100 frames/sec

Frame duration = $\frac{1}{100}$

= 0.01 sec

= 10 ms

Bit rate = 32 x 100

= 3200 bps

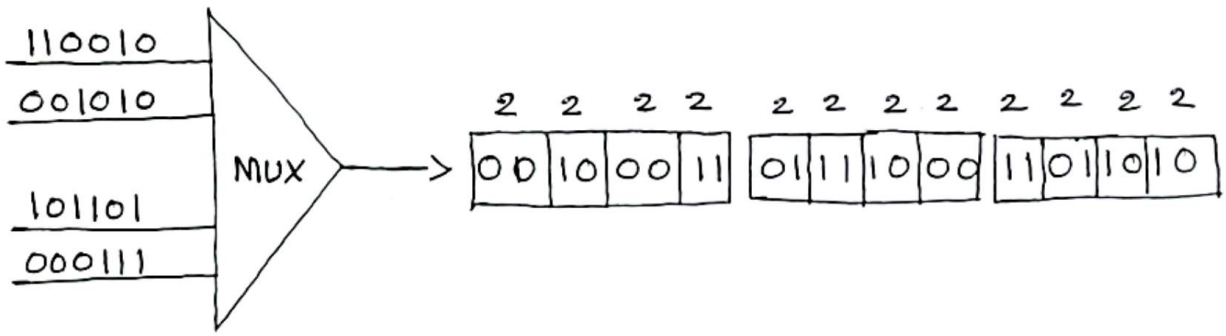
Channels = 4
Each sends = 100 bytes

Ex - 6.9

A multiplexer combines four 100-kbps channels using a time slot of 2 bits. Show the output with 4 arbitrary inputs. What is the frame rate? What is the frame duration? What is the bit rate? What is the bit duration?

=>

Output structure -



Frame size = 3 x 2 = 6 bits

a) Frame rate = $\frac{100 \text{ kbps}}{2} = \frac{100000}{2} = 50000$ frames/sec

b) Frame duration = $\frac{1}{50000} = 20 \mu s$

c) Bit rate = 4 x 100 kbps = 400 kbps

d) Bit duration = $\frac{1}{400000} = 2.5 \mu s$

Error Detection

Error in communication is a condition when the received data/information doesn't match with the transmitted data/information.

Types of error -

1. Single bit error -

When only one bit of transmitted data is altered.

Sent = 10<u>1</u>10 ; Received = 10<u>0</u>10

2. Multiple bit error -

When more than one bit of transmitted data is altered.

Sent = 1<u>0</u>1<u>1</u>0 ; Received = 1<u>1</u>1<u>0</u>0

3. Burst error -

When two or more consecutive bits of transmitted data is altered.

Sent = 1<u>011</u>0 ; Received = 1<u>100</u>0

Error detection methods -

1. Simple parity check
2. Two-dimensional parity check
3. Checksum
4. Cyclic Redundancy check (CRC)

Error correction -

1. Forward error correction
2. Backward error correction

—o—

1. Simple parity check -

In this technique an extra bit, known as parity bit is added to data unit before transmission.

*** —> Even parity :-

If ~~total~~ number of 1's in the original data bit is odd then an extra '1' is added at the end.

Ex: 1011 —> 1011 1

1001 —> 1001 0

—> Odd parity :-

If number of 1's in the original databit is even then an extra '1' is added added at end.

Ex: 1011 —> 1011 0

1001 —> 1001 1

2. Two dimensional parity check :-

Here parity bits are calculated for each row and column, similar to a simple parity check.

Ex- Original data,

1101 1000 1010

Now,

	Row
1101	1
1000	1
1010	0
Column → 1111	0

← Check even / odd

Sent data,

11011 10001 10100 11110

(Not much important)

Same math using odd →

1101	0
1000	0
1010	1
0000	1

Sent data,

11010 10000 10101 00001

3. Checksum :-

In this method, the sender divides the data into fixed-size segments and computes a total sum of them and does 1's complement. The computed checksum is transmitted along with the data. At the receiver, the same computation is performed to verify data was not corrupted.

Ex Original data,

10011001 11100010 00100100 10000100

Sender

```
  10011001
  11100010
  00100100
(+)10000100
-----------
 10|00100011
 ↓ carry
   (+) 10
-----------
  00100101
-----------
  11011010 -> Checksum
↓ Compliment
```

Receiver,

```
  10011001
  11100010
  00100100
  10000100
  11011010
-----------
 10|11101101
   (+) 10
-----------
  11101110
-----------
  00010001
↓ Compliment
```

Sent data,

10011001 11100010 00100100 10000100 11011010

4. Cyclic Redundancy Check -

This method uses polynomial division to verify the integrity of data during transmission or storage.

Ex

Let a message, 11100 that is to be sent over a medium and the divisor is given polynomial x^3+1. Find error through CRC method.

=>

Original data = 11100

Divisor = $x^3+1 = 1 \times x^3 + 0 \times x^2 + 0 \times x^1 + 1 \times x^0$

$= 1\ 0\ 0\ 1$

Extra bit = No. of divisor bit - 1

$= 4 - 1 = 3$

extra bit

```
1001 ) 11100 000
       1001
       ----
       01110
        1001
        ----
        01110
         1001
         ----
         01110
          1001
          ----
          0111 0
           1001
           ----
           0111   -> Remainder
```

{XOR function of data & divisor}

P.T.O

Now,

received data -

```
  11100000
      111
  --------
  11000111   -> Transmitted data.
```

Now transmitted data error check,

```
1001 ) 11100111
       1001
       ----
       0111
        1001
        ----
        0110
         1001
         ----
         0101
          1001
          ----
          0100
           1001
           ----
           0000   -> No error,
```

So data accepted.

```
* 1101
  1001
  ----
  0100
```

Here,

If same data in up and down.

It will be = 0.

If different = 1.

Hamming code -

Hamming code is a set of error correction codes that can be used to detect and correct errors that occur during transmission.

Formula- $2^P \geq m + P + 1$

m = data bit

P = Parity bit = 0, 1, 2, 3, ...

-o-

Imagine given data = 1011

=> So, $m = 04$

$P = 0, 1, 2, 3, \ldots$

$1 = 2^0 \geq 4 + 0 + 1 = 5$

$2^1 = 2 \geq 4 + 1 + 1 = 6$

$2^2 = 4 \geq 4 + 2 + 1 = 7$

$2^3 = 8 \geq 4 + 3 + 1 = 8$

So total parity bit, $P = 3$ -> $2^0 = 1 = P_1$

$2^1 = 2 = P_2$

$2^2 = 4 = P_4$

So block,

Number of block = $m + P = 4 + 3 = 7$

D_7	D_6	D_5	P_4	D_3	P_2	P_1

(D = Data bit)

=>

1	0	1	P_4	1	P_2	P_1

Now,

$P_1 = D_3\ D_5\ D_7$ → checks 1 bit, skips 1 bit. (including P_1)

$= 1\ 1\ 1$ → For even pair (3 → 1 = Odd)

$\therefore P_1 = 1$

$P_2 = D_3\ D_6\ D_7 = > 0$ → checks 2 bit including P_2,

$= 1\ 0\ 1$ skips 2 bit.

$P_4 = D_5\ D_6\ D_7 = > 0$ → checks 4 bit including P_4,

$= 1\ 0\ 1$ skips 4 bit.

$\therefore$ Transmitted data,

D_7	D_6	D_5	P_4	D_3	P_2	P_1
1	0	1	0	1	0	1

= ~~Ans~~

=>

Let error message,

1	0	1	0	0	0	1
D_7	D_6	D_5	P_4	D_3	P_2	P_1

$\therefore P_1 = \underline{1} => D_3\ D_5\ D_7 = 011 \rightarrow P_1 = \underline{1}$ } If error, then 1.

$P_2 = \underline{0} => D_3\ D_6\ D_7 = 001 \rightarrow P_2 = \underline{1}$ }

$P_4 = \underline{0} => D_5\ D_6\ D_7 = 101 \rightarrow P_4 = \underline{0}$ } If ok, then 0.

$\therefore P_4\ P_2\ P_1 => 011 => 3$ (Decimal value) = 3rd bit issue.

$\therefore$ Corrected bit =

1	0	1	0	1	0	1

Ans

Prob 1

Cyclic redundancy check:

Sent data = 100100 and key = 1101; after ~~that~~ error at receiver, data = 100000. Find remainder and check error at receiver end.

=>

Sent data = 100100

key = 1101

∴ Extra bit = 3

Now,

```
1101 ) 100100000
       1101
       ----
       01000
        1101
        ----
        01010
         1101
         ----
         01110
          1101
          ----
          00 1100
             1101
             ----
             0001
```

So final remainder = 001 (Final remainder bit will be equal to the number of extra bit)

Ans

∴ Transmitted data

= 100100000 + 001 = 100100001

(P.T.O)

Now,

received data = 100000

CRC extra bit

∴ received data bit = 100000 000 + 001

= 100000001

```
1101 ) 100000001
       1101
       ----
        1010
        1101
        ----
         1110
         1101
         ----
          1100
          1101
          ----
           011   -> Remainder.
```

So, Error found at receiver end.

Ans

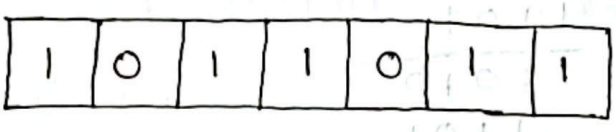
Prob 2

Hamming code:

Received data = 1011011 ; check for error and correct it.

=>

Received error data,



$2^P \geq m + P + 1$

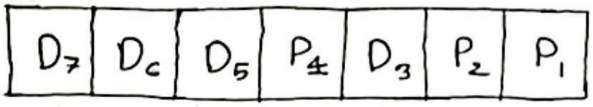
$=> 2^3 \geq 4 + 3 + 1 = 8$

$\therefore m = 4$

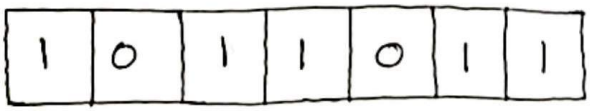
$P = 3 \rightarrow 2^0 = 1$

$2^1 = 2$

$2^2 = 4$



=>



$P_1 = 1 \rightarrow D_3\ D_5\ D_7 = 011 \rightarrow P_1 = 1$

$P_2 = 1 \rightarrow D_3\ D_6\ D_7 = 001 \rightarrow P_2 = 0$

$P_4 = 1 \rightarrow D_5\ D_6\ D_7 = 101 \rightarrow P_4 = 1$

$P_4 P_2 P_1 = 101 = 5$ (Decimal)

$\therefore$ Correct code ->



Prob 3

Hamming code:

For given data find the parity bits. Data = 10011.

=>

Data bit, $m = 5$

$\therefore 2^p \geq m + p + 1$

$=> 2^0 = 1 \geq 5 + 0 + 1 = 6$

$=> 2^1 = 2 \geq 5 + 1 + 1 = 7$

$=> 2^2 = 4 \geq 5 + 1 + 1 = 8$

$=> 2^3 = 8 \geq 5 + 3 + 1 = 9$

$=> 2^4 = 16 \geq 5 + 4 + 1 = 10$

$\therefore p = 4$; $2^0 = 1 = P_1$

$2^1 = 2 = P_2$

$2^2 = 4 = P_4$

$2^3 = 8 = P_8$

No. of block $= m + p = 5 + 4 = 9$

D_9	P_8	D_7	D_6	D_5	P_4	D_3	P_2	P_1

=>

1	P_8	0	0	1	P_4	1	P_2	P_1

$\therefore P_1 = D_3\ D_5\ D_7\ D_9 = 1001 \rightarrow P_1 = 1$

$P_2 = D_3\ D_6\ D_7 = 100 \rightarrow P_2 = 1$

$P_4 = D_5\ D_6\ D_7 = 100 \rightarrow P_4 = 1$

$P_8 = D_9 \Rightarrow 1 \rightarrow P_8 = 1$

P.T.O

∴ Transmitted data,



Ans

Prob 4

Data = 11001 10000 10101 11011, show how checksum works for given data.

=>

Sender,

```
  11001
  10000
  10101
  11011
-------
10|11001
   10
  ----
   11011
  -----
   00100  -> Checksum
```

Check error,

```
  11001
  10000
  10101
  11011
  00100
-------
10|41101
   10
  ----
   11111   } compliment.
  -----
   00000
```

No error.

Sent data,->

11001 10000 10101 11011 00100

Radio Receiver

AM Radio Receiver –

A device which reproduces the modulated or radio waves into sound waves.

Receiver action :-

1. Receive the passing radio waves.
2. Select the right signal by passing through LC circuit.
3. Must amplify the received signal.
4. Retrive the audio signal from amplified radio wave.
5. Amplify the audio signal and feed to speaker.

AM Radio Receiver types –

→ Straight Radio Receiver

→ Superhytrodyne Receiver

Straight Radio Receiver –

→ Receiver intercepts radio waves from different broadcasting stations.

→ The radio ~~receiver~~ wave is amplified by the tuned RF amplifier.

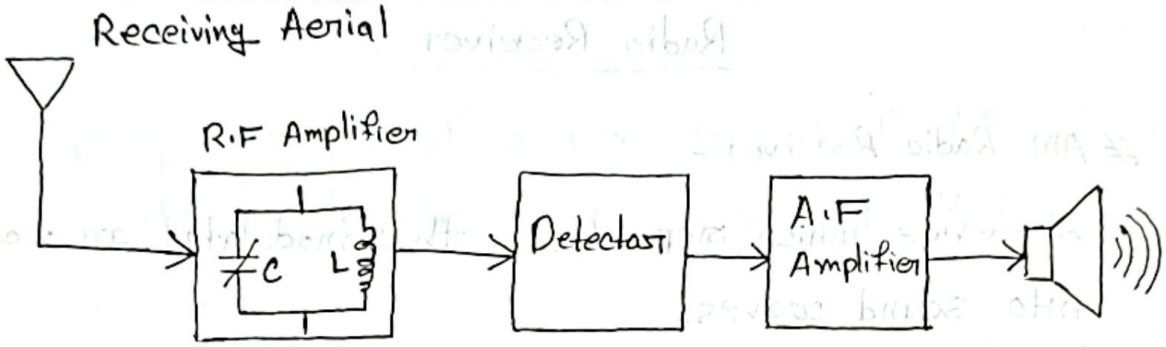
→ Detector circuit extracts audio signal from radio wave.

→ Output goes through audio amplification.

→ At last amplified audio signal is fed to the speaker.

→ Limitation –

Circuit needs to be tuned for desired station.



Superheterodyne Receiver :-

—> R.F Amplifier circuit-

Selects the desired radio signal and amplifies it using a tuned circuit.

—> Mixer stage -

Combines the RF signal with a local oscillator to produce an intermediate frequency (IF).

—> IF Amplifier-

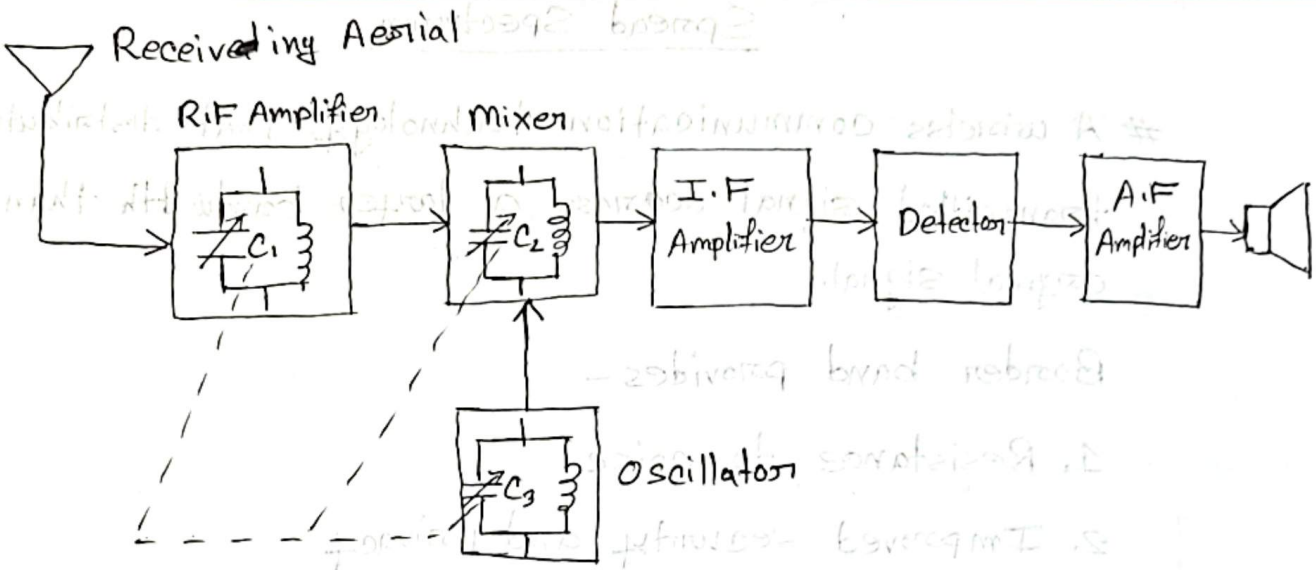
Amplifies the intermediate frequency signal for further processing.

—> Detector stage-

Extracts the audio signal from the IF signal.

—> AF Amplifier-

Amplifies the audio signal to drive the speaker and produce sound.



Advantages-

1. The superhetrodyne principle makes it possible to produce an IF which is much less than the RF.
2. Decreases loss compared to straight receiver.
3. Permits use of fixed RF Amplifiers.
4. Better noise rejection.
5. Bandwidth depends on IF, so reception & same audition quality for all stations.

Spread Spectrum

A wireless communication technology that distributes the transmitted signal across a larger bandwidth than the orginal signal.

Border band provides-

1. Resistance to noise
2. Improved security and privacy
3. Multiple user access
4. High noise immunity
5. Immune to jamming



Types-

1. Frequency Hopping Spread Spectrum (FHSS)
2. Direct Sequence Spread Spectrum (DSSS)

Pseudo Noise Sequence –

A coded sequence of 1s & 0s with certain auto-correlation properties called as Pseudo-Noise code.

–> A deterministic binary sequence generated by a known algorithm.

–> Helps detect error in transmitted signal at receiver end.

#

–> Spread spectrum method is used in Code Division Multiple Access (CDMA).

–> In FDMA & TDMA-

–> Bandwidth is limited

–> Low spectral efficiency

–> Narrow band. easily affected by noise.

–> Spread Spectrum -

–> Has interference immunity

–> Low power density

–> High bit rate

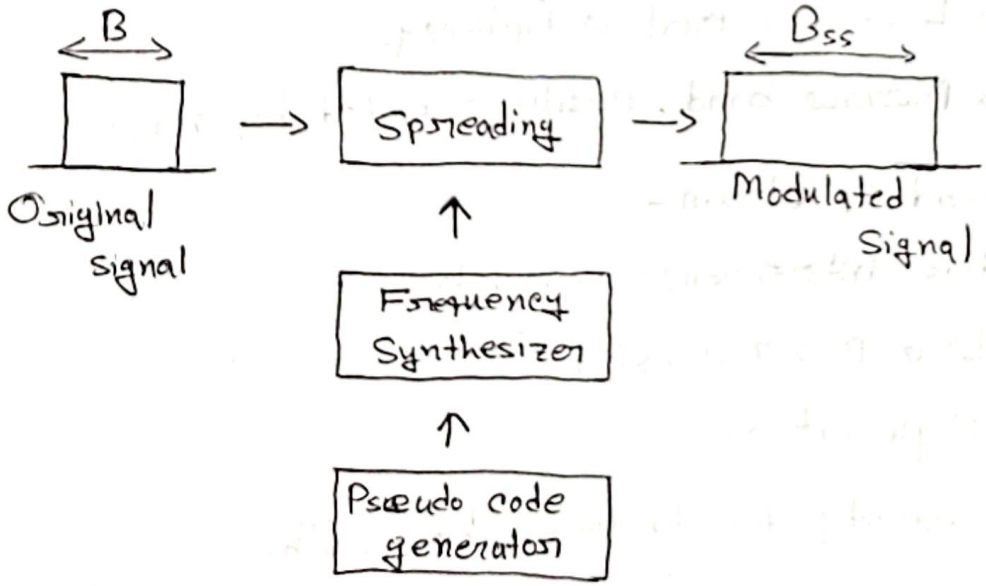
–> Security due to larger bandwidth.

FHSS-

In this technique, the carrier frequency rapidly switches among many frequency channels according to a pseudo-random sequence known to both transmitter and receiver.

-> A frequency was alloted to sender 1 for a particular period of time. After a while, sender 1 hops to the other frequency and sender 2 uses the first frequency. This is called frequency reuse.

-> The amount of time spent on each frequency hop is called Dwell time.



Application-

Bluetooth, Military Communication, Walkie-Talkie, Wireless LAN.

Advantages -

1. Sender doesn't need to wait
2. Signal has high strength
3. Never affected by interference
4. Affordable & commonly used
5. Processing gain is higher than DSSS.

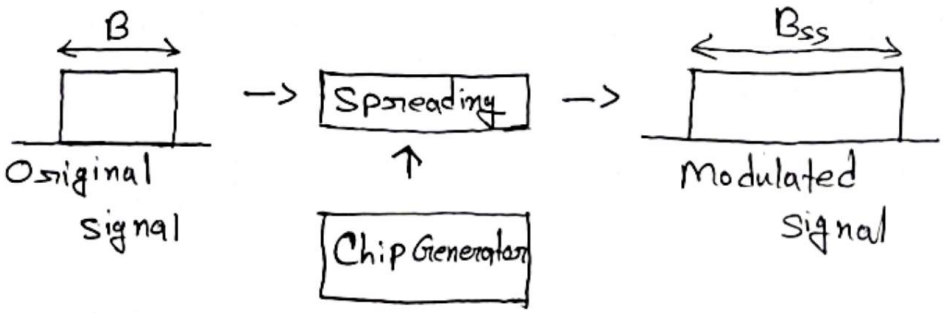
Disadvantages -

1. Hard to find user frequency at given time.
2. Requires precise synchronization.
3. Complex frequency control.

DSSS -

It is a spreading technique in which the data signal is multiplied by a high-rate pseudo-noise code to spread it over a much wider frequency band.

-> Each & every bit of the user data is multiplied by a secret code, called as chipping-code.



Applications:

GPS, Cellular network, Satellite communication.

Advantages-

1. Most effective solution to Jamming.
2. Performance in presence of noise is superior to FHSS.
3. Minimizes signal interference

Disadvantages-

1. Signal power strength is low.
2. Expensive compared to FHSS.
3. Complex receiver.